METRIC

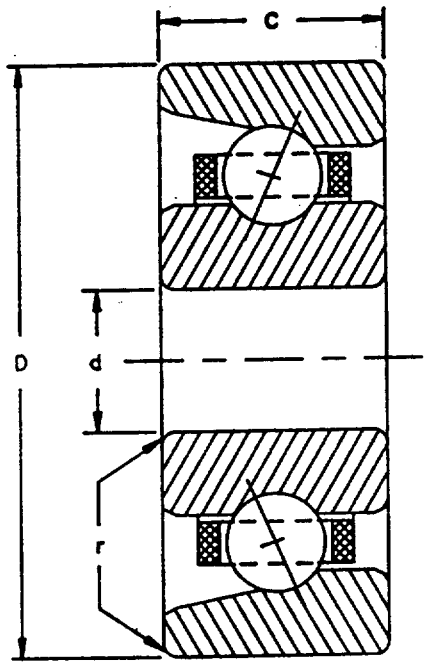
MIL-B-913/9
5 March 1993

MILITARY SPECIFICATION SHEET

BEARING, BALL, ANNULAR, FOR INSTRUMENTS AND PRECISION ROTATING COMPONENTS, ANGULAR CONTACT, UNFLANGED, NON-SEPARABLE AND COUNTERBORED OUTER RING, CHROMIUM-ALLOY STEEL, ABEC 5P (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-913.



BEARING CONFIGURATION

AMSC N/A

FSC 3110

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REQUIREMENTS:

1. Design. All bearings described by this specification sheet shall be angular contact instrument bearings with an unflanged, non-separable and counterbored outer ring confifuration.

2. Material. The ball and ring material for these bearings shall be chromium-alloy steel 52100 (UNS G52986) conforming to AMS 6444.

3. Tolerance class. The tolerance class shall be in accordance with the ABEC 5P tolerance tables of ANSI/AFBMA Standard 12.1. This tolerance class shall apply to all bearing sizes listed in Table I of this specification sheet.

4. Boundary dimensions. The boundary dimensions shall be as specified by the size code in Table I.

5. Retainer. The retainer shall be as specified by the part number designator (PN DES) in Table II.

6. Closures. Angular contact bearings shall be supplied with no closures.

7. Radial internal clearance or contact angle. The radial internal clearance or contact angle shall be as specified by the part number designator in Table IV.

8. Calibration. Calibration shall be as specified by the part number designator in Table V.

9. Performance Test. The performance test shall be as specified by the part number designator in Table III.

10. Lubrication.

10.1 Lubricant. The lubricant shall be in accordance with the specification specified by the part number designator in Table VI.

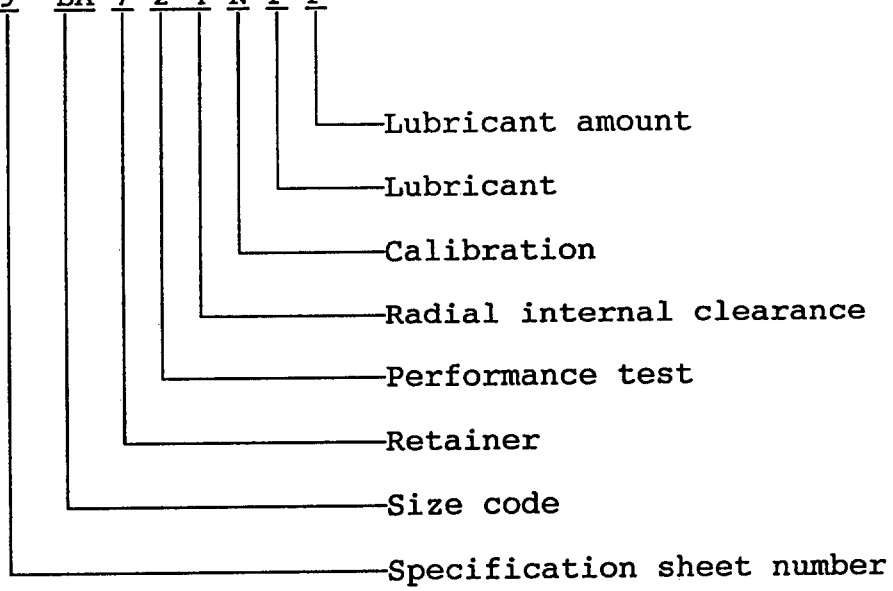
10.2 Lubricant amount. The amount of lubricant shall be as specified by the part number designator in Table VII.

10.3. Barrier coating. Barrier coating shall be applied to bearings as specified by the part number designator in Table VI.

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11. Part number. The part number consists of the follwing: Prefix letter M; general specification number; number of this specification sheet; and alpha numeric codes for size, retainer, performance test, radial internal clearance or contact angle, calibration, lubrication, and amount of lubricant.

EXAMPLE: M 913/ 09 BA 7 2 4 N P P



M913/09BA724NPP indicates - Bore 2.0 mm; outside diameter 5.0 mm; width 1.5 mm; radius 0.08 mm; retainer, phenolic laminate; contact angle, greater than 14° but less than 20°; no calibration; performance test, starting torque; lubricant, preservative; lubricant amount, in accordance with MIL-P-197.

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TABLE I. Boundary dimensions (millimeters).

SIZE CODE	BORE	OD	WIDTH	RADIUS
	d	D	C	r 1/ 2/
AA	1.5	5.0	2.0	0.15
BA	2.0	5.0	1.5	0.08
BB	2.0	6.0	2.3	0.08
CA	2.5	6.0	1.8	0.08
CB	2.5	7.0	2.5	0.15
DA	3.0	7.0	2.0	0.10
DB	3.0	10.0	4.0	0.15
EA	4.0	9.0	2.5	0.10
EB	4.0	13.0	5.0	0.20
EC	4.0	16.0	5.0	0.30
FA	5.0	11.0	3.0	0.15
FB	5.0	16.0	5.0	0.30
FC	5.0	19.0	6.0	0.30
GA	6.0	13.0	3.5	0.15
GB	6.0	19.0	6.0	0.30
HA	7.0	14.0	3.5	0.15
HB	7.0	19.0	6.0	0.30
HC	7.0	22.0	7.0	0.30
JA	8.0	19.0	6.0	0.30
JB	8.0	22.0	7.0	0.30
KA	9.0	20.0	6.0	0.30
KB	9.0	26.0	8.0	0.30
LA	10.0	26.0	8.0	0.30

1/ Maximum shaft or housing fillet radius that bearings corners will clear.
2/ Narrow face of rings shall clear a maximum fillet radius of one-half the value listed. This value shall be a minimum of 0.05.

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TABLE II. Retainer.

PN DES	TYPE
6	PTFE TOROIDS 1/
7	PHENOLIC LAMINATE 2/
8	VACUUM IMPREGNATED PHENOLIC 3/
9	NON-POROUS, NON-METALLIC CROWN

1/ PTFE (Polytetrafluoroethylene).
2/ Phenolic or other porous non-metallic material.
3/ Phenolic or other porous non-metallic material saturated with lubricant. Selection of this choice dictates choosing manufacturer's standard amount of lubricant.

TABLE III. Performance Test.

PN DES	TYPE
1	MANUFACTURER'S STANDARD
2	STARTING TORQUE 1/

1/ Starting torque limits form table III of MIL-B-913.

TABLE IV. Radial internal clearance or contact angle.

PN DES	RANGES
1	2µm to 7µm (micrometers)
2	5µm to 12µm
3	14° OR LESS 1/
4	GREATER THAN 14° BUT LESS THAN 20° 1/
5	12µm to 20µm
6	GREATER THAN 20° 1/
8	20µm to 28µm

1/ Angular contact bearing designations only.

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TABLE V. Calibration of bore and outside diameter (micrometers).

PN DES	BORE INCREMENTS	OD INCREMENTS
N	NO CALIBRATION	NO CALIBRATION
A	NO CALIBRATION	2.50
B	NO CALIBRATION	1.25
C	2.50	NO CALIBRATION
D	2.50	2.50
E	2.50	1.25
F	1.25	NO CALIBRATION
G	1.25	2.50
H	1.25	1.25

TABLE VI. Lubricant.

PN DES	SPECIFICATION
P	PRESERVATIVE 1/
A	MIL-L-6085
B	MIL-L-6085 2/
C	MIL-L-81846
D	MIL-L-81846 2/
E	MIL-G-23827
F	MIL-G-81322
G	MIL-G-81937
H	MIL-G-83261
J	MIL-S-81087
K	MIL-S-81087 2/

1/ PN DES "P" shall be used only with PN DES "P" in TABLE VII.
2/ With barrier coat.

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TABLE VII. Lubricant amount.

PN DES	QUANTITY
P	PRESERVATIVE AMOUNT AS REQUIRED BY MIL-P-197
1	MANUFACTURER'S STANDARD 1/
2	OIL PER TABLE II of MIL-B-913
3	15% GREASE PER TABLE II of MIL-B-913
4	25% GREASE PER TABLE II of MIL-B-913
5	35% GREASE PER TABLE II of MIL-B-913
6	45% GREASE PER TABLE II of MIL-B-913

1/ Grease, fill to minimum 25%, maximum 40% of bearing void. Oil, immerse and then allow excess to drip off. The standard quantity of oil varies with each bearing size.

Custodians:
Army - AT
Navy - AS
Air Force - 99

Review activities:
Navy - SH
DLA - IS

User activities:
Navy - MC
Air Force - 84

Preparing activity:
Navy - AS

Agent:
DLS - IS

(Project 3110-0813-09)

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